

# ECE 612: Applied Machine Learning Engineering

## Term: Summer 2025

### Administrative and Instructor Information

#### Course Prerequisite

Strong familiarity with programming in python.

#### Course Meeting Time and Location

Friday 11:00 am-1:50pm in Curtis Hall 250A.

#### Course Staff

**Instructor:** Somayeh Keshavarz

Office hours: TR 2:15 -PM 3:15 PM Bossone 102.

Email: [sk4388@drexel.edu](mailto:sk4388@drexel.edu) *include ECE612 in the subject line of all course-related emails.*

Other meetings by email appointment.

### Student Learning Information

#### Course Description

This course emphasizes how to gather data then train, test, and deploy practical machine learning systems using modern software libraries, with an emphasis on Keras on TensorFlow. Complementing the other department courses emphasizing the mathematics behind machine learning algorithms and the ways these can be tailored to specific computing architectures, this project-focused course emphasizes the practice of rapidly prototyping and testing multiple learning structures. To provide the broadest applicability, datasets will range from rich text, to financial time series, to sound, images, video, and control.

#### Course Purpose with a Plan of Study

Complementing other department courses emphasizing the mathematics behind machine learning algorithms, and the ways these can be tailored to specific computing architectures, this project-focused course emphasizes the practice of rapidly prototyping and testing multiple learning structures.

#### Statement of Expected Learning

1. The ability to gather and ingest data, select appropriate loss functions and learning architectures, then test and train them, all in a modern machine learning library.
2. Familiarity with the commonly used learning architectures for various types of data (text, time series, images, audio, video), and how to select leading models and implement transfer learning with them in a modern machine learning library.
3. Experience with optimizing learning algorithm and structure parameters to particular problems and datasets.
4. Experience with training learning controllers through game play.
5. Experience with visualizing information gleaned from data through machine learning.

## Course Materials

All of the texts below are available online via Drexel's library. Reading will be assigned from the first text.

- *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems, 3rd Ed.*, Aurélien Géron. O'Reilly Media, Inc. Sebastopol, CA. ISBN 9781098122478. Available from Drexel's library at <http://ezproxy2.library.drexel.edu/login?url=https://learning.oreilly.com/library/view/~/9781098125967/?ar&orpq&email=~u>.
- *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems, 2nd Ed.*, Aurélien Géron. O'Reilly Media, Inc. Sebastopol, CA. ISBN 9781492032649. Available online via Drexel's library at [https://drexel.primo.exlibrisgroup.com/permalink/01DRXU\\_INST/dbok2j/alma991013509424804721](https://drexel.primo.exlibrisgroup.com/permalink/01DRXU_INST/dbok2j/alma991013509424804721)
- *Deep Learning with Python, 2nd Ed.*, Francois Chollet. Manning Publications, Shelter Island, NY. ISBN 9781617294433. Available online via Drexel's Library at [https://drexel.primo.exlibrisgroup.com/permalink/01DRXU\\_INST/dbok2j/alma991017069585904721](https://drexel.primo.exlibrisgroup.com/permalink/01DRXU_INST/dbok2j/alma991017069585904721)

Additionally, each lecture will be accompanied with lecture slides, which serve as the primary resource for the course.

## Course Website and Online Materials

You will need to check the course website periodically to obtain assignments and lecture materials. We will use the BBLearn system to deliver all information regarding the course, including project descriptions, links to online materials, and grades as they are calculated. Your projects are to be submitted through this system in a manner consistent with each project description. However, emails should not be sent through this system, but rather directly to the instructor at the email addresses above.

## Course Schedule and Assignments

### Required Course Reading

Each week you should read the chapters listed below from *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 3rd Ed.* Assignments are due according to the Graded Assignments section above.

| week | required reading                                                                               |
|------|------------------------------------------------------------------------------------------------|
| 1    | Chapters 1 - 4 (ML Taxonomy, Intro. to Classification, Linear Regression, Logistic Regression) |
| 2    | Chapters 10 - 11 (Neural Networks with Keras & Optimizers)                                     |
| 3    | Chapters 12 - 13 (Data in TensorFlow)                                                          |
| 4    | Chapter 14 (CNNs and Images)                                                                   |
| 5    | Chapter 14, continued (Transfer Learning)                                                      |
| 6    | Chapter 17 (Generative Models)                                                                 |
| 7    | Chapter 18 (Reinforcement Learning)                                                            |
| 8    | Chapter 15 (Sequences, Time Series, & RNNs)                                                    |
| 9    | Chapter 16 (NLP with RNNs & Attention)                                                         |
| 10   | Chapter 19 (Deploying Models at Scale)                                                         |
| 11   | Exam week. Final Assignment due (see Graded Assignments above). No final exam.                 |

## Course Grading Policy (Assignments and Assessments)

### Graded Assignments

As this course is a practicum, the course grade will be entirely determined by grades on 5 machine learning projects. The criteria for assessment of each project will be provided in the project description. The majority of the project grades will be dictated by the loss obtained by the provided trained learning structure on data yet un-revealed to the student. The lowest project grade will be dropped, and the

average of the remaining 4 projects will be used to obtain a final score. This score will be assigned to a letter grade as indicated in the table to the right.

| week | project subject                            |  |
|------|--------------------------------------------|--|
| 2    | Basic Keras Model Workflow                 |  |
| 4    | Transfer Learning for Image Classification |  |
| 6    | Generative Modeling                        |  |
| 8    | Learning to Control with TensorFlow Agents |  |
| 10   | Working with Text                          |  |

## Grading Matrix

Each project will receive a score out of 100 following the scoring rubric described in the project's description. At least 25 % of this scoring rubric will be computed by ranking the loss obtained by the student's provided trained learning structure on data yet un-revealed to the student.

*The lowest project score will be dropped, then the remaining 4 project grades will be averaged (equally weighted – i.e. 25% each), to obtain a final course score.*

## Grade Scale

The course letter grade will be assigned based on the final course score via the following ranges. A: 95-100, A-:90-94, B+:87-89, B:83-86, B-:80-82, C+:77-79, C:73-76, C-:70-72, D+:65-69, D:60-64, F:0-59.

At my discretion, I may curve course grades up, but a student will never receive a grade lower than the one computed as described above.

## University Academic Policies

**Course Change Policy:** The instructor reserves the right to modify the course, as necessary, during the term: including policies, evaluations, due dates, course content, schedule, assignments or requirements. All changes will be communicated in lecture and/or via the course DrexelLearn page.

*Please review the following relevant university policies.*

### Academic Integrity, Plagiarism, Dishonesty and Cheating Policy:

<https://drexel.edu/provost/policies/academic-integrity/>

**Disability Resources:** Students requesting accommodations due to a disability at Drexel University need to request a current Accommodations Verification Letter (AVL) in the ClockWork database before accommodations can be made. These requests are received by Disability Resources (DR), who then issues the AVL to the appropriate contacts. For additional information, visit the DR website at <http://drexel.edu/oed/disabilityResources/overview/>, or contact DR for more information by phone at 215.895.1401, or by email at [disability@drexel.edu](mailto:disability@drexel.edu).

**Course Add/Drop Policy:** <http://www.drexel.edu/provost/policies/course-add-drop>

**Course Withdrawal Policy:** <http://drexel.edu/provost/policies/course-withdrawal>

**Weather, Emergencies and University Closing:** University closing or delayed opening information will be posted on <http://www.drexel.edu>. In the event of the need to close or delay the daily opening of a campus, the University will provide notice via Web, telephone, and the DrexelALERT system. Closing or delayed opening information will be announced at 215-895-MELT (6358). The University determines whether to close or delay opening due to inclement weather, not the instructor. Therefore, please do not contact the instructor for this information.